RHAN Evolution & Epoch Analysis Report

A Detailed Evaluation of Training Durations, Architectural Additions, and Epoch Sufficiency across the RHAN Lineage

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Executive Summary

The Recurrent Hybrid Attention Network (RHAN) series represents a systematic effort to narrow the human-AI visual robustness gap. Through 13 distinct iterations, we have integrated biological priors—recurrent feedback, ventral/dorsal pathway separation, frequency filtering, and semantic anchoring.

An essential finding of our research is that **architectural complexity and training schedule must be co-calibrated**. Several promising architectures were prematurely refuted or underperformed because their non-linear optimization landscapes (e.g., dynamic gating, VAE reconstruction priors, and self-supervised contrastive learning) were not allocated sufficient epochs to converge.

This report catalogs all RHAN versions, specifies their training schedules, and identifies which configurations require extended training to truly validate or refute their underlying hypotheses.

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1. Lineage, Epochs, and Feature Catalog

Version / Variant	Dataset	Total Epochs	Key Additions & Features	Current Status / Robustness Metric
RHAN-clean	CIFAR-10	80–100	Baseline ConvStem + Transformer + Recurrent Feedback; Clean CE loss.	**Complete:** ε_{thresh} = 0.0330
RHAN-adv	CIFAR-10	80–100	Baseline architecture; Standard PGD-5 training (ε = 0.031).	**Complete:** ε_{thresh} = 0.0764
RHAN-v3	CIFAR-10	100	Ventral/Dorsal pathway split (256-dim each); Neural alignment to CORnet-S IT on adversarial images.	**Complete:** ε_{thresh} = 0.0900
RHAN-v4	CIFAR-10	100	Multi-scale feedback (4 scales); Active CLIP semantic loss during adversarial training; InfoNCE	**Complete (Refuted):** ε_{thresh} = 0.0800. Suffered from Semantic-Geometric

			consistency.	Conflict.
RHAN-v5	CIFAR-10	150	Learnable frequency separation; Phase-decoupled pretraining (Phase 0: 30 ep clean CLIP; Phase 1: 120 ep PGD curriculum).	**Complete (Validated):** ε_{thresh} = 0.1030. Spontaneous M-pathway shape dominance.
RHAN-v5-TRADES	CIFAR-10	120	Replaced custom loss with standard TRADES (β = 6.0) + IT alignment. Initialized from CLIP pretraining.	**Complete:** ε_{thresh} = 0.1113
RHAN-TRADES-Hardened	CIFAR-10	120+	Class-hardened ε scaling (up to 0.055 for vulnerable classes); 20-step adaptive APGD; inter-class margin loss.	**Complete:** ε_{thresh} = 0.1246. AutoAttack (AA) on car/truck collapsed to 0%.
RHAN-trades-curriculum	CIFAR-10	60	3 phases × 20 epochs (ε = 0.062 → 0.100 → 0.150) starting from the hardened checkpoint.	**Complete:** ε_{thresh} = 0.1850. Ceiling for 32×32 CIFAR; car/truck AA still 0%.
RHAN-v6	CIFAR-10	120	Input-dependent $\alpha(x)$ frequency gating; predictive coding feedback; Adaptive Computation Time (ACT).	**Regressed (Under-optimized):** Clean 82.03%, no meaningful ε_{thresh}.
RHAN-v7	CIFAR-10	90	VAE encoder/decoder heads (latent_dim=256); generative classifier; fresh random perceptual critic; online feature comparison.	**In Progress (Under-optimized):** VAE pixel reconstruction conflicts with TRADES.
RHAN-v8	CIFAR-10	120	CLIP text feature anchoring to prevent representation drift under curriculum expansion.	**Complete:** Prevented representation drift.
			Self-Supervised Invariance	**Concept**

RHAN-v9 (SAIL)	CIFAR-10	130	Pretraining (SAIL) via InfoNCE on clean/adv pairs (50 ep) + TRADES (60 ep) + Concept tuning (20 ep).	**Concept Design (Under-optimized):** Pretraining phase too short.
RHAN-TDV (Clean)	STL-10	100	Phase 0 TDV pretraining (30 ep) on unlabeled STL-10; Phase 1 Label Calibration (10 ep); Phase 2 Clean TRADES Consistency (60 ep).	**Complete (Refuted):** Clean 78.5%, AA 1.76% (car/truck collapse to 0%).
RHAN-TDV (Adversarial)	STL-10	100	Same as above, but Phase 2 enforces Adversarial TDV Consistency ($z_{adv}[t] + m_t = z_{clean}[t+1]$).	**Complete (Under-optimized):** Clean 75.4%, AA 0.78%. Capacity bottleneck.
RHAN-Pseudolabel-TRADES	STL-10	168	Phase 0 (30 ep CutMix); Phase 1-8 TRADES curriculum (138 ep) on combined 5K labeled + 22K high-confidence pseudo-labels.	**Complete:** Clean 86.20%, ε_{thresh} = 0.0130. Car class collapsed to 0% under AA.
RHAN-UNIFIED	STL-10	138	WideSEConvStem (~20.5M params); 3-layer transformer; cosine head; CutMix; 3-ep β warmup at phase transitions.	**In Progress:** Phase 0 best = 72.94%; Phase 1 & 2 stable.

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2. In-Depth Analysis: Which Models Need More Epochs?

RHAN-v6 (Dynamic Gating & Adaptive Computation Time)

- **Hypothesis:** Dynamic gating ($\alpha(x)$) and ACT allow the network to adjust its processing depth and frequency filtering dynamically based on the noise level of each input.
- **Why it was refuted/regressed:** During the 6-phase curriculum (120 epochs total, or only 20 epochs per phase), the model collapsed in the final phase (ε = 0.250). ACT maxed out its pondering budget, and the dynamic gates failed to stabilize.
- **Scientific Rationale for More Epochs:** The inclusion of ACT introduces a discrete or highly non-linear halting policy, while dynamic gating parameterizes the input pipeline. The joint optimization landscape of a transformer, dynamic gating networks, and ACT is highly non-convex. Training for only 20 epochs per

phase does not allow the policy network to learn a smooth halting threshold, causing it to fall back to the maximum step limit (max pondering).

- **Recommended Training Schedule:**
  - **Increase to 300 epochs** (50 epochs per phase).
  - Introduce an explicit entropy regularizer on the ACT halting probability to encourage exploration of shorter paths early in training.
  - Reduce the learning rate by 2× for the gating parameters to allow the main backbone weights to stabilize first.

🟣 RHAN-v7 (Generative World-Model)

- **Hypothesis:** A generative prior (VAE decoder) forces the network to only map inputs to latent representations that can be reconstructed into plausible images, creating a “manifold constraint” that blocks adversarial perturbations.
- **Why it is underperforming/in-progress:** Pixel-level reconstruction objectives conflict with TRADES, and training is 3× slower (~900s/epoch). Currently, Phase 0 VAE pretraining is restricted to 30 epochs.
- **Scientific Rationale for More Epochs:** VAE decoders trained on complex distributions require hundreds of epochs to generate sharp, structurally coherent reconstructions. With only 30 epochs of Phase 0 training, the decoder outputs blurry images. Adversarial attacks easily exploit these fuzzy reconstruction boundaries because the manifold constraint is too weak.
- **Recommended Training Schedule:**
  - **Extend Phase 0 VAE Warmup to 150 epochs** to guarantee the decoder learns sharp object boundaries.
  - **Extend Phase 1 Joint Training to 120 epochs** to allow the classifier and the VAE decoder to co-adapt smoothly without gradient conflicts.

🟣 RHAN-v9 (SAIL & True Predictive Coding)

- **Hypothesis:** Self-Supervised Adversarial Invariance Learning (SAIL) pretrains the backbone representations to be invariant ($f(x_{adv}) \approx f(x_{clean})$) prior to classification, while Predictive Coding feedback replaces raw feature feedback with prediction error residuals.
- **Why it is under-optimized:** The SAIL pretraining phase is currently allocated only 50 epochs.
- **Scientific Rationale for More Epochs:** Self-supervised contrastive learning (InfoNCE) lacks the strong gradient signals of supervised cross-entropy. Standard contrastive frameworks (like SimCLR, MoCo, or VICReg) require 200–800 epochs to reach representation stability. When combined with adversarial perturbations, the network needs even longer to map the perturbed ε-ball back to the clean representation anchor. 50 epochs is insufficient; the representation space remains unaligned.
- **Recommended Training Schedule:**
  - **Increase SAIL Pretraining (Phase 1) to 200 epochs** at a higher batch size.
  - Maintain the 60-epoch TRADES fine-tuning phase, which will converge rapidly once the underlying representations are robustly invariant.

🟣 RHAN-TDV (Temporal Difference in Vision)

- **Hypothesis:** Restricting representations using the video-causal constraint ($z_t + m_t = z_{t+1}$) prevents feature collapse and builds temporal invariants that block high-frequency noise.
- **Why it underperformed:** Under AutoAttack ($\varepsilon = 0.031$), overall robustness fell to 0.78% (Adversarial) and 1.76% (Clean), despite strong clean accuracy

(78.5%).
- **Scientific Rationale for More Epochs:** Phase 0 video pretraining was conducted for only 30 epochs on UCF-101. Self-supervised video models (like DINO-temp or VIBCREG) require hundreds of epochs to learn motion-static decoupling. With only 30 epochs, the motion encoder and the image encoder learn local, high-frequency texture correlations (shortcuts) to satisfy the temporal difference equation, rather than genuine shape-based dynamics. Under adversarial attack, these texture shortcuts are immediately destroyed.
- **Recommended Training Schedule:**
  - **Extend Phase 0 Video Pretraining to 150–200 epochs** on UCF-101.
  - Maintain LayerNorm and the raw feature variance penalty to ensure feature space standard deviation remains high ($Std \geq 0.45$).

RHAN-Pseudolabel-TRADES (STL-10)

- **Hypothesis:** Using 22,322 high-confidence pseudo-labels mined from the 100K unlabeled STL-10 images provides the data diversity needed to stabilize the decision boundaries of the 20.5M parameter WideSEConvStem.
- **Why it is under-optimized:** Despite boosting clean accuracy to 86.20% and the $d' = 1.0$ threshold to $\varepsilon = 0.0130$, the model collapsed to 0% on the `car` class under AutoAttack at $\varepsilon = 0.031$.
- **Scientific Rationale for More Epochs:** Pseudo-labels are inherently noisy. According to deep learning memorization dynamics (Arpit et al., 2017), networks learn simple, clean patterns during the early stages of training and memorize noisy/incorrect labels during the later stages. With only 15 epochs per curriculum phase (138 epochs total), the model does not have enough epochs to filter out pseudo-label noise and generalize a smooth decision boundary, leading to local boundary vulnerabilities that AutoAttack exploits.
- **Recommended Training Schedule:**
  - **Extend curriculum phases to 25 epochs each (230 epochs total)**.
  - Incorporate a cosine learning rate decay scheduler with a longer tail to allow the network to slowly minimize the noise contribution of the pseudo-labels.

RHAN-UNIFIED (STL-10 96×96)

- **Hypothesis:** Higher-resolution training (96×96 vs 32×32) provides 9× more pixels, enabling true shape discrimination and reducing the relative impact of $\varepsilon = 0.031$ noise (perturbing only ~1% of image info instead of ~9%).
- **Why it needs more epochs:** The WideSEConvStem increases parameters to 20.5M, making it highly susceptible to overfitting or representation collapse under a fast curriculum.
- **Scientific Rationale for More Epochs:** The current configuration limits phases 5–8 to 12 epochs each. At high epsilon levels ($\varepsilon \geq 0.094$), the Squeeze-and-Excitation (SE) blocks need to learn complex, dynamic channel gain controls to suppress noise. 12 epochs is too short; the optimizer does not have time to recover from the learning rate and epsilon transitions, causing representation drift.
- **Recommended Training Schedule:**
  - **Increase Phases 5–8 from 12 epochs to 25 epochs each**.
  - Keep the 3-epoch beta warmup at the start of each phase transition to prevent optimizer shock.

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3. Recommended Architectural & Training Epoch Adjustments

Based on these findings, we recommend the following modifications to our active training configurations:

```
gantt
    title Recommended Epoch Extensions for Future Runs
    dateFormat  X
    axisFormat %d

    section VAE (v7)
    Phase 0 VAE Warmup (Old: 30)      :active, 0, 30
    Phase 0 VAE Warmup (Extended)   :crit, 30, 150
    Phase 1 Joint Training (Extended):crit, 150, 270

    section SAIL (v9)
    Phase 1 SAIL Pretrain (Old: 50)  :active, 0, 50
    Phase 1 SAIL Pretrain (Extended) :crit, 50, 250
    Phase 2 TRADES Fine-tuning       :250, 310

    section TDV (STL-10)
    Phase 0 Video Pretrain (Old: 30) :active, 0, 30
    Phase 0 Video Pretrain (Extended):crit, 30, 180
    Phase 2 TRADES Curriculum        :180, 240
```

Actionable Next Steps:

1. **For the current STL-10 TDV training run:** Increase the `--epochs` flag in the script for Phase 0 from `30` to `150`.
2. **For the Unified STL-10 run:** Modify the curriculum schedule in `train_rhan_unified.py` to increase the later phases (Phases 5-8) to `25` epochs each.
3. **For the Generative World-Model (v7):** Separate the VAE training into a dedicated `train_vae_pretrain.py` script that runs for `150` epochs to ensure high-quality reconstructions before beginning classification.